Colton Quirk, B.S., MSc

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Education

University of St Andrews, School of Physics and Astronomy St Andrews, Scotland
Master's of Science (MSc) in Astrophysics with Merit September 2025 — August 2026

- Relevant Coursework: AS5525 Advanced Extragalactic Astronomy, AS5004 Cosmology, AS5011 General Relativity, AS5524 Astrophysical Fluid Dynamics, AS5523 Gravitational Dynamics and Accretion Physics
- Second Semester Research: *Dark Matter in Early Type Galaxies*
  - Modeled the dark matter content of NGC 2764 using Jeans Anisotropic Modeling.
  - Developed a pipeline for the analysis of the dark matter content in a sample of galaxies.
- Thesis: *Kinematic Signatures of Spiral Shocks*
  - Developed a Julia package, `SPHAnalysis`, to analyze Smoothed Particle Hydrodynamic (SPH) galaxy simulation data.
  - Analyzed data from a SPH galaxy simulation to identify signatures of shocking regions within spiral arms of galaxies.

University of Arizona, Honors College, College of Science Tucson, AZ
Bachelor's of Science (B.S.) in Physics and Astronomy | Minor: Computer Science August 2021 — May 2025

- GPA: 3.98/4.0 | Dean's List with Distinction, Galileo Circle Scholar (Department of Physics), Phi Beta Kappa
- Relevant Coursework: ASTR 302 Introduction to Observational Astronomy, ASTR 300A/B Astronomy and Astrophysics, ASTR 400A/B Theoretical Astrophysics, PHYS 305 Computational Physics
- Honors Thesis: *Scientific Visualization of Photon Ring Formation and Imaging*
  - Developed a custom rendering engine for Blender to view models around a Schwarzschild black hole.
  - Created a scientifically accurate movie demonstrating the imaging of a black hole based on the BHEX mission.

Work Experience

Research Internship May 2024 — August 2024
National Radio Astronomy Observatory (NRAO), National Science Foundation REU Socorro, NM

- *Probing Radio Environments of Superclusters in the Very Large Array Sky Survey (VLASS)*
  - Statistical analysis of brightest cluster galaxies (BCGs) within galaxy clusters associated with superclusters and field BCGs not associated with superclusters.
  - Found a link between power output, P3GHz, and redshift for all of the galaxies observed and a link between higher power output for BCGs within the core of superclusters compared to field and non-core BCGs.

Research Internship September 2023 — May 2024
Arizona Space Grant Consortium, University of Arizona Tucson, AZ

- *Revealing the Far-Ultraviolet Background Radiation through Archival Astronomical Data*
  - Analyzed archival FIMS/SPEAR data to construct a far-ultraviolet background radiation map.

Telescope Operator September 2022 — May 2025
Steward Observatory, University of Arizona Tucson, AZ

- Operated the Raymond E. White, Jr. 21″ telescope for weekly observatory nights and special functions.
- Guided ~40 students per night through celestial observations, explained astronomical concepts, and responded to inquiries regarding telescope functionality and operations.

Telescope Operator Summers 2021 — 2023
Montana Learning Center (MLC) Helena, MT

- Operated the 25″ Newtonian Dobsonian telescope, at the time, the largest public-use telescope in Montana.
- Remotely operated MLC's 16″ research-grade Ritchey-Chretien telescope located in New Mexico.
- Star Parties: Engaged with ~100 attendees per event, selected celestial objects for viewing, delivered informational talks on astronomical phenomena, and addressed audience questions.

Skills

- **Programming Languages**: Julia, Python, Javascript, HTML/CSS, Java, Fortran
- **Technologies**: Git, Linux, LaTeX, Quarto, Typst, CASA, Blender, NumPy, SciPy, Astropy, Matplotlib, XEphem
- **Languages**: English, Spanish